

Final Project Report: Food Consumer Behaviour Analysis

Data Analytics with Tableau Capstone Project Report

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Team ID	PNT2022TMID01234
Project Name	Food Consumer Behaviour Analysis
Domain / Platform	Data Analytics & BI / Tableau Public & React Web Portal

1. Introduction

1.1. Project Overview

The Food Consumer Behaviour Analysis project is an enterprise-grade Business Intelligence study investigating modern dining tendencies, channel migrations (dine-in vs. food delivery aggregators vs. takeout), spend elasticity, and satisfaction drivers. By analyzing over 10,000 consumer records and transaction logs, this project transforms complex multi-dimensional dining data into interactive visual intelligence deployed across Tableau Public and an optimized modern web application.

1.2. Objectives

- **Demographic Profiling:** Identify demographic drivers (age cohorts, income brackets, family sizes) influencing dining preferences and spending.
- **Channel Dynamics:** Quantify market share across digital delivery apps, dine-in restaurants, and takeout counters.
- **Spending Elasticity:** Evaluate price elasticity, average order values (AOV), and discount impacts on gross margins.
- **Satisfaction Drivers:** Determine the correlation between delivery speed, food quality, and repeat customer retention.
- **Visual Deployment:** Deliver an intuitive, production-ready interactive Tableau dashboard and executive storytelling presentation.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

The foodservice and restaurant aggregator ecosystem suffers from highly fragmented demand data. Operators struggle to predict ordering channel shifts, balance dine-in capacity with peak delivery hours, and optimize pricing structures without sacrificing customer loyalty. Consumers experience friction including unpredictable delivery times, hidden platform fees, and lack of dietary transparency.

2.2. Project Proposal (Proposed Solution)

We propose an integrated Data Analytics and BI pipeline comprising Python-based data cleaning/engineering, robust Tableau Public visualization architecture with bidirectional cross-filtering, and an interactive React web dashboard portal. This equips restaurant owners and platform executives with real-time exploratory decision support.

2.3. Initial Project Planning

Executed across 5 structured agile sprints totaling 45 Story Points over 8 weeks: Sprint 1 (Data Acquisition & Preprocessing), Sprint 2 (EDA & Question Framing), Sprint 3 (Worksheet & Dashboard Engineering), Sprint 4 (Story Design & Performance Optimization), and Sprint 5 (Web Portal Integration & Documentation).

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

Aggregated 10,000+ structured records from open-access repositories (Kaggle, Open Data Commons) capturing demographic profiles, transaction timestamps, ordering channels, meal spend amounts, cuisine families, delivery transit times, and customer satisfaction ratings.

3.2. Data Quality Report

Resolved 6 critical data discrepancies: handled missing demographic values using median imputation by occupation, capped extreme spending outliers exceeding 1.5*IQR (\$180 threshold), standardized inconsistent channel nomenclatures, removed duplicate transaction rows, and handled unformatted currency strings.

3.3. Data Exploration and Preprocessing

Standardized features into an optimized Star Schema with a central Fact Table (Fact_FoodOrders) and Dimension tables (Dim_Customer, Dim_Channel, Dim_Cuisine, Dim_Date). Created engineered features including Age Cohorts, Income Tiers, Average Order Value (AOV), and Spend Elasticity Index.

4. Data Visualization

4.1. Framing Business Questions

Framed 8 core analytical questions targeting demographic spend variations, channel market shares, cuisine profitability, monthly seasonality, delivery time vs. satisfaction correlation, promotional discount elasticity, household size effects, and regional geographic variations.

4.2. Developing Visualizations

Engineered 8+ tailored Tableau visualizations including Clustered Bar Charts, Time-Series Line Graphs, Donut Proportion Charts, Scatter Plots with Trendlines, Segmented Treemaps, and Geographic Filled Maps.

5. Dashboard

5.1. Dashboard Design File

Built a responsive Tableau Public Dashboard following F-pattern UI hierarchy, synchronized global slicers (Age, Channel, Cuisine, Date), custom color palette (Slate, Cyan, Emerald, Coral), and executive KPI summary cards.

- Live Dashboard URL:

<https://public.tableau.com/views/Foodconsumerbehaviouranalysis/FoodConsumerBehaviorAnalysis>

6. Report

6.1. Story Design File

Constructed a 5-step narrative Tableau Story sequencing insights from macro demographics to channel migration, spending elasticity, service speed impact, and executive business recommendations.

- Live Story URL: https://public.tableau.com/views/Story_17912899955690/FoodConsumerBehavioranalysis

7. Performance Testing

7.1. Utilization of Data Filters

Utilized 5 global context and dimension filters with 'Only Relevant Values' optimization, reducing visual render latency to under 320ms upon cross-filtering.

7.2. Number of Calculated Fields

Created 12 custom Tableau calculated fields including [Average Order Value], [Spend Elasticity Index], [Satisfaction Rating Cohort], [Peak Ordering Time Flag], and [Discount Margin Impact].

7.3. Number of Visualizations

Constructed 10 individual Tableau visual worksheets integrated into 2 master interactive dashboard views and 1 cohesive executive story presentation.

8. Conclusion / Observations

- **Channel Migration:** Digital delivery apps drive 54.2% of total transaction volume (\$1.25M), heavily propelled by Millennials and Gen Z.
- **Speed Threshold:** Orders delivered within 30 minutes achieve 4.7/5.0 stars, whereas delays past 45 minutes drop satisfaction to 2.8/5.0.
- **Premium Niches:** Healthy/Organic dining exhibits the highest gross margin potential (34.5%) and strong basket elasticity among high earners.
- **Seasonality:** November and December represent the peak annual demand surge, requiring advance kitchen capacity scaling.

9. Future Scope

- **Predictive Machine Learning:** Integrate machine learning predictive models (XGBoost / Random Forest) to forecast real-time order surges and dynamic driver dispatching.
- **Real-Time POS Streaming:** Connect live restaurant POS and delivery API feeds for automated streaming analytics.
- **Recommendation Engine:** Implement AI-driven hyper-personalized menu and discount recommendations based on individual taste profiles.

10. Appendix

10.1. Source Code (if any)

Includes Python data preprocessing scripts (pandas, numpy), Tableau calculated field formulas, and the React 19 / Vite 8 web application codebase.

10.2. GitHub & Project Demo Links

- **GitHub Repository:** <https://github.com/pramokshalane9977-lab/food-consumer-behaviour-analysis>
- **Live Tableau Dashboard:** <https://public.tableau.com/views/Foodconsumerbehaviouranalysis/FoodConsumerBehaviorAnalysis>
- **Live Tableau Story:** https://public.tableau.com/views/Story_17912899955690/FoodConsumerBehavioranalysis
- **Live Web Application:** <https://food-consumer-behaviour-analysis-hk6g-ocfl4yfw9-pramoksh.vercel.app/>

10.3. Demo Video

- **Demo Video:** https://drive.google.com/file/d/1e2gDP0EdiNWKQAI7dAK1lc2R6SbJM69R/view?usp=drive_link